Can MLLMs Reason in Multimodality? EMMA: An Enhanced MultiModal ReAsoning Benchmark

Yunzhuo Hao, Jiawei Gu, Huichen Will Wang, Linjie Li, Zhengyuan Yang, Lijuan Wang, Yu Cheng

UESTC, Sun Yat-sen University, University of Washington, Microsoft, CUHK

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Instructions

- This talk summarizes the EMMA benchmark and findings.
- Slides use concise bullet points; see figures and tables for details.
- Key terms are highlighted with color.

Motivation

- Humans solve problems by combining textual and visual reasoning.
- Current benchmarks often allow text-only shortcuts.
- Question: Can MLLMs perform organic multimodal reasoning requiring back-and-forth between modalities?
- We introduce EMMA to focus on tasks where vision is integral, not decorative.

Example of text-extracted visual info: $y = 0.5^x$ can be read off an image without real graphical reasoning.

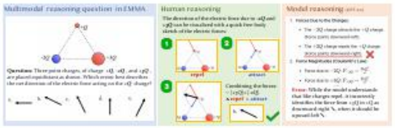
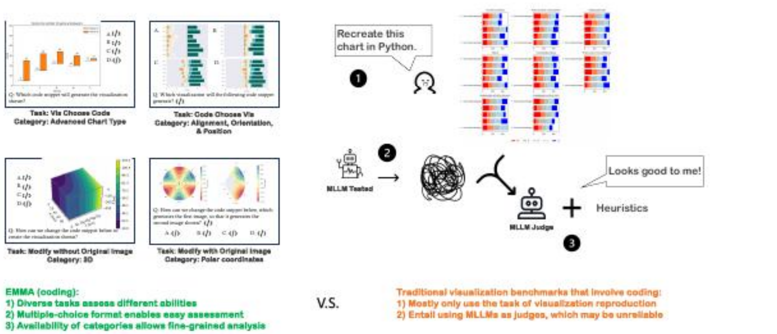


Figure 1: A representative EMMA physics item stressing visual reasoning.

Related Work

- Multimodal benchmarks: many test surface perception or text-dominant reasoning.
- Some filtering (e.g., removing text-solvable items) exists, but often still permits single-pass visual perception to suffice.
- EMMA advances prior work by:
  - Filtering out items solvable from text + image captions.
  - Adding fine-grained skill labels and many new items requiring visual simulation/transformation.



Method: Benchmark Design and Filtering Pipeline

- Goal: retain items that truly require multimodal reasoning.
- Filtering:
  - Generate captions for images (GPT-4o).
  - Give text + captions to strong models (multiple attempts).
  - Discard if any model gets $\geq 5/10$ correct.
- Remaining items need visual simulation, spatial transformation, or multi-hop reasoning.

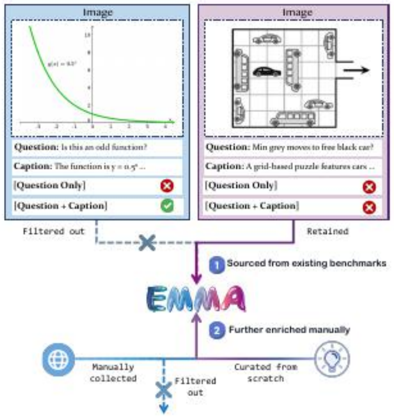


Figure 3: EMMA's stricter filtering: removes text- or caption-solvable items.

Method: Composition of EMMA

- 2,788 questions, across Math, Physics, Chemistry, Coding.
- 1,796 are newly created; the rest curated and filtered from existing sets.
- Multiple-choice and short open-ended answers; no MLLM judge needed.
- Fine-grained skill labels enable category-level analysis.

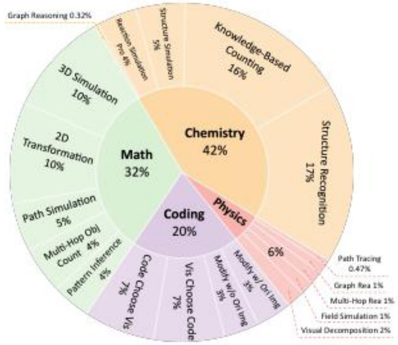
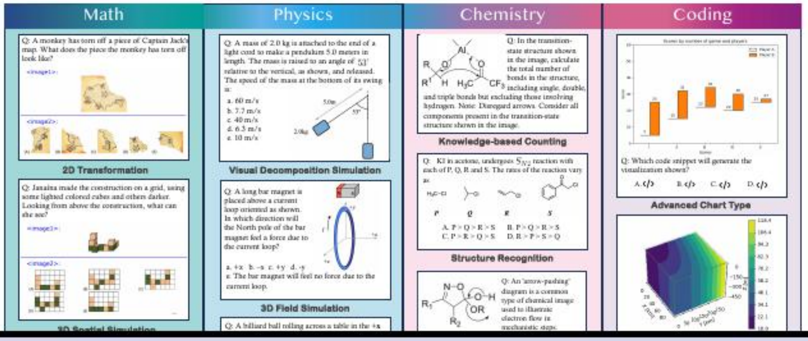


Figure 4: Composition and labeling overview of EMMA.

Method: Subject Taxonomies

- Math: 2D Transformation, 3D Spatial Simulation, Path Tracing, Multi-Hop Object Counting, Pattern Inference.
- Physics: 3D Field Simulation, Graph Reasoning, Multi-Hop Visual Reasoning, Path Tracing, Visual Decomposition.
- Chemistry: Counting, Structure Recognition, Reaction Simulation, Reaction Simulation Pro, Graph Reasoning.
- Coding: Vis Choose Code, Code Choose Vis, Modify (with/without original image); fine-grained design labels.



Method: Examples Emphasizing Visual Reasoning

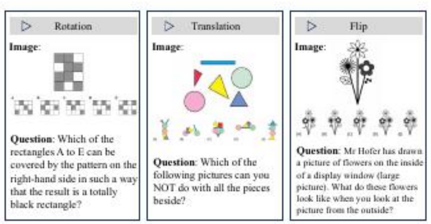


Figure 6: Math 2D transformations require mental rotation/flip/translation.

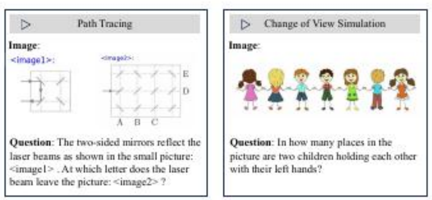
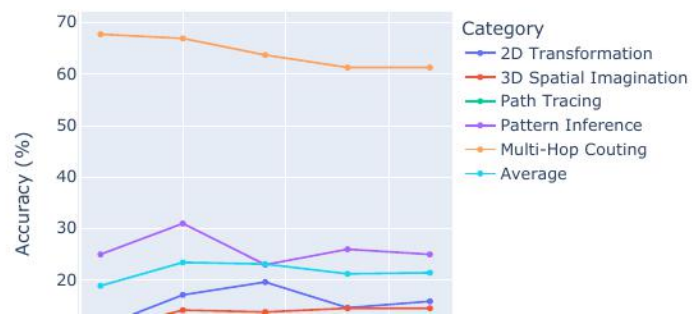


Figure 7: Path tracing and view changes need multi-step visual simulation.

Experimental Method

- Two prompting strategies:
  - Direct: output answer only.
  - Chain-of-Thought (CoT): step-by-step reasoning text.
- Test-time compute scaling:
  - Majority Voting, Best-of-N (BoN), Tournament-Style.
  - Reward model variants: Self, Gemini 2.0 Flash Thinking, specialized math RM.
- Balanced subset EMMA-mini (100 per subject) for constrained APIs.



Experimental Setting

- Models evaluated (zero-shot):
  - Closed-source: GPT-4o, Claude 3.5 Sonnet, Gemini 2.0 Flash, Gemini 2.0 Flash Thinking, o1 (mini only).
  - Open-source: Qwen2-VL-72B-Instruct, LLaVA-OneVision-72B, InternVL2-Llama3-76B, InternVL2.5-78B.
- Inference:
  - Default parameters unless noted; vLLM for fast pass@n where applicable.
  - Answers: MCQ or short text in a box format in original evaluation.
- Datasets:
  - EMMA full benchmark.
  - EMMA-mini (balanced 400) for head-to-head comparisons.

Experimental Results: EMMA-mini Overall

- All models are far from human performance.
- Best model on EMMA-mini overall: o1 at 45.75%.
- Gemini 2.0 Flash Thinking reaches 43.50%.
- Human experts: 77.75% average.

System	EMMA-m
Human Experts	
o1	
Gemini 2.0 Flash Thinking	
Qwen2-VL-72B-Instruct (Direct)	
GPT-4o (Direct)	

Table 1: Condensed overall results on EMMA-mini (from main results).

Ablation: CoT Effects and Error Patterns

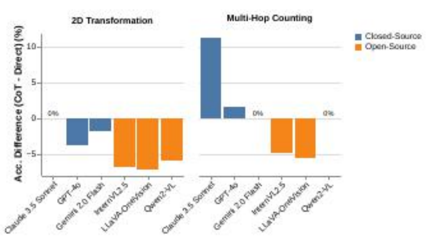


Figure 9: CoT helps some tasks, can hurt visual-centric ones.

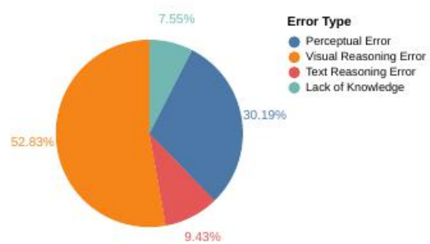
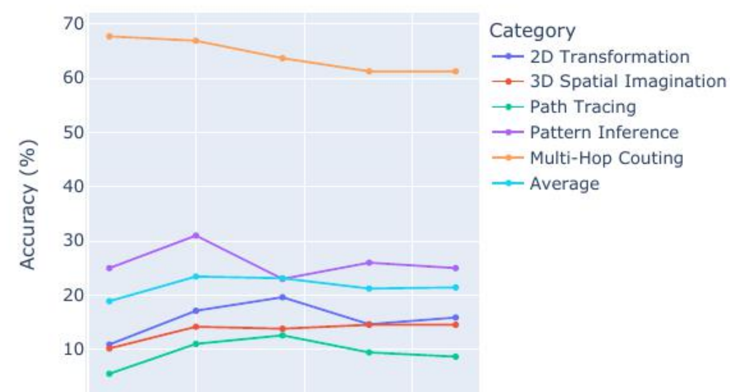


Figure 10: Dominant errors are visual reasoning failures.

- Visual simulation/spatial errors dominate; textual CoT cannot replace missing visual CoT.
- CoT generally boosts closed-source, but often hurts open-source models.

Ablation: Test-time Scaling and Reward Models

- Majority voting, BoN, tournament improve performance modestly (e.g., +7.25 pp).
- Stronger reward models (e.g., Gemini 2.0 Flash Thinking) outperform self-RM.
- Specialized text-only math RMs are less reliable on multimodal math.



Deficiencies and Limitations

- Physics portion is smaller than other subjects due to sourcing difficulty.
- Chemistry currently emphasizes organic tasks; broader coverage desirable.
- Visual reasoning remains a bottleneck—current MLLMs often skip essential simulations.
- Textual CoT alone is insufficient for strongly visual tasks.

Future Research Directions

- Architectures enabling visual Chain-of-Thought (explicit visual steps).
- Better reward models for multimodal evaluation and selection.
- Expanded domains (more physics); richer multi-image and 3D reasoning.
- Training paradigms emphasizing visual simulation and spatial cognition.
- Practical applications: debugging visualizations, STEM tutoring, design assistance.

Thank You

Questions?

Slides summarized from the EMMA paper.